Manage access

Assign capabilities

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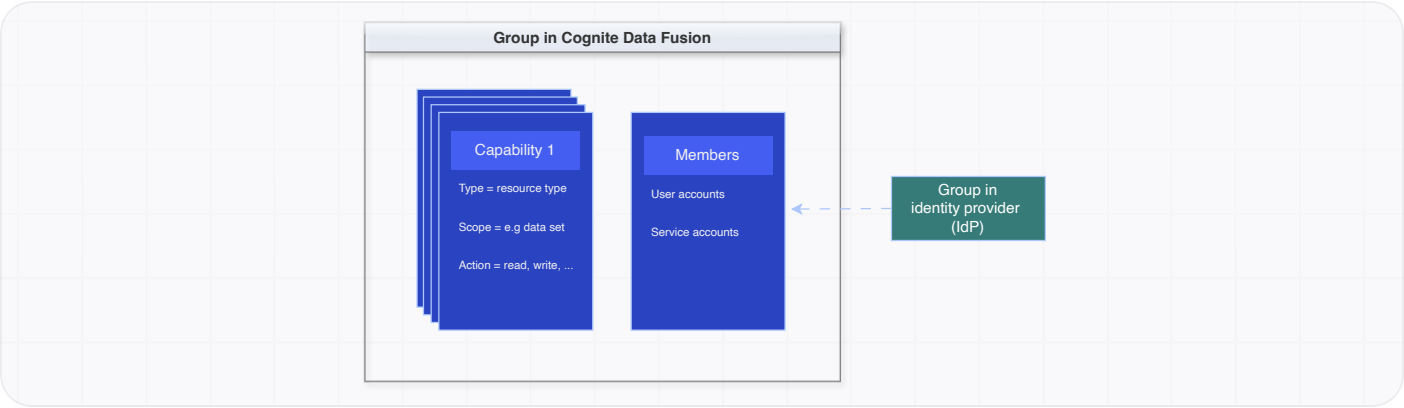
Control access to data and features in Cognite Data Fusion (CDF) by assigning capabilities to groups.

To control access to data and features in Cognite Data Fusion (CDF), you define what **capabilities** users or applications have to work with different **resource types** in CDF, for example, if they can *read* a *time series* (`timeseries:read`) or *create* a *3D model* (`3D:create`).

Capabilities also decide which **features** you have access to. For example, you need the `3d:create` capability to upload 3D models to CDF.

A capability is defined by a **resource type**, a **scope**, and **actions**. The resource type and scope define **the data** the capability applies to, while the action defines **the operations** you are allowed to perform.



Instead of assigning capabilities to individual users and applications, you use **groups** in CDF to define which capabilities the group members (users or applications) should have. You **link** and synchronize the CDF groups to user groups in your identity provider (IdP), for instance, Microsoft Entra ID or Amazon Cognito.

You can tag sensitive resources with additional **security categories** for even more fine-grained access control and protection.

This flexibility lets you manage and update your data governance policies quickly and securely. You can continue to manage users and applications in your organization's IdP service outside of CDF.

This article explains how to **add capabilities to groups**, **create and assign security categories**. You will also find overviews of the necessary **capabilities to access and use different features in CDF**.

> For users to successfully sign in and use Fusion UI, the following minimum set of capabilities is required : `projects:list` , `groups:list` and `groups:read`

Create a group and add capabilities

1. **Navigate to the Groups page**

   Navigate to **Admin** > **Groups** > **Create group**.

2. **Enter a name**

   Enter a unique name for the group.

3. **Add capabilities**

   - In the **Capability type** field, select a resource type, such as assets and time series, CDF groups, data sets, or specific functionality.

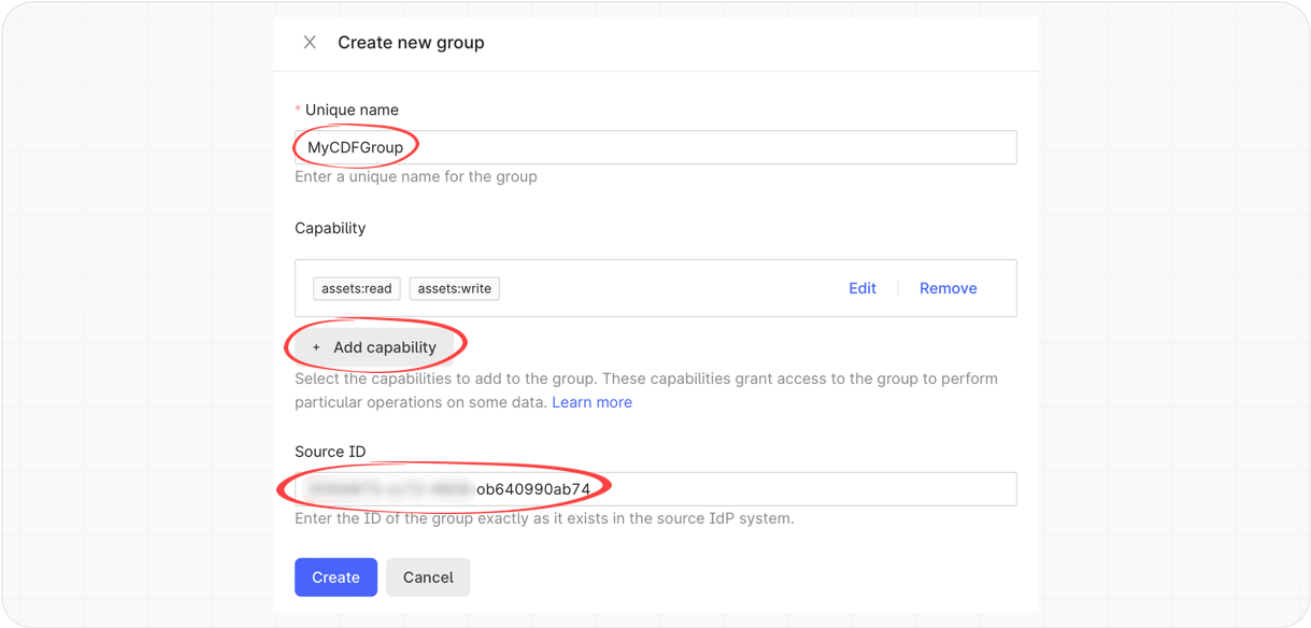
includes all data as an option.

4 **Save the group**

Select **Save**.

5 **Link the group to your identity provider**

In the **Source ID** field, enter the **Object Id** (Entra ID) or **Group name** (Cognito) exactly as it exists in your identity provider (IdP). It will link the CDF group to a group in Microsoft Entra ID or Amazon Cognito.



Create and assign security categories

You can add an extra access level for **time series** and **files** by tagging resources with **security categories** via the Cognite API. This is useful if you want to protect market-sensitive data. To access resources tagged with a security category, you must have both the standard capabilities for the resource type and capabilities for the security category.

- `securitycategories:update`
- `securitycategories:delete`

To assign security categories to groups:

1. **Open the group**

   Open the group where you want to add security categories.

2. **Select Security categories**

   In the **Capability type** field, select **Security categories**.

3. **Set the security category**

   In the **Action** field, select `securitycategories:memberof`.

4. **Set the scope**

   In the **Scope** field, select **Security categories**, associate a security category or select **All**.

To perform actions, such as `read` or `write` on **time series** and **files** tagged with capabilities and security categories:

- You must be a member of a group with actions that give access to a times series or files, for instance, `timeseries:read`.
- You must be a member of a group with the `securitycategories:memberof` capability for the same time series or files.

All users with **any** group membership in a CDF project get the `userProfilesAcl:READ` capability and can search for other users.

Feature capabilities

The tables below describe the necessary capabilities to access different CDF features.

In addition to the capabilities listed in the sections below, users and applications need these **minimum** capabilities to access any feature in CDF.

Capability type	Action	Scope	Description
Groups	`groups:list`	Current user, All	Verifies user group access.
Projects	`projects:list`	All	Verifies that a user or application has access to a CDF project. To access the resources in the project, see the capabilities listed below.

Atlas AI

Build and deploy AI agents to automate tasks and workflows.

Capability type	Action	Scope	Description
Agents	`agents:read`, `agents:write`, `agents:run`	All	Build agents.
Agents	`agents:run`, `agents:read`	All	Interact with agents.

instances	data retrieval.

You may need more capabilities depending on the tools used by the agent.

Canvas

Add assets, engineering diagrams, sensor data, images, and 3D models to a canvas.

To set up a canvas and its features, you'll need **extended access**. To work on a canvas, you'll need **end-user access**.

Extended access

Users with extended access can set up the **Canvas** tool, create **threshold rules**, and set up **canvas labels**:

		`pace`	**Canvas** tool. - `datamodels:write` is required to set up the canvas labels feature.
Data model instances	`datamodelinstances:read`, `datamodelinstances:write`	`cdf_industrial_canvas`, `cdf_apps_shared`, `IndustrialCanvasInstanceSpace`, `SolutionTagsInstanceSpace`, `RuleInstanceSpace`	- Scope `SolutionTagsInstanceSpace` to create, edit, and delete canvas labels. - Scope `RuleInstanceSpace` to set up threshold rules.

End-user access

Add these capabilities for users that will work on a **canvas**:

Capability type	Action	Scope	Description
Data models	`datamodels:read`	`cdf_industrial_canvas`, `cdf_apps_shared`, `IndustrialCanvasInstanceSpace`, `CommentInstanceSpace`	- Scope `IndustrialCanvasInstanceSpace` to use the Canvas tool. - Scope `CommentInstanceSpace` to use the Comment feature on a canvas.
Data model instances	`datamodelinstances:read`	`cdf_industrial_canvas`, `cdf_apps_shared`, `IndustrialCanvasInstanceSpace`, `CommentInstanceSpace`, `SolutionTagsInstanceSpace`, `RuleInstanceSpace`	- View, create, update, delete canvas data. - Scope `CommentInstanceSpace` to use the Comment feature on a canvas. - Scope

Resource types	`read` actions for the resource types you want to add to canvas	Data sets, All	View assets, events, files, time series, and 3D models
Files	`write`	Data sets or All	Add **local files** on a canvas. This is an optional capability.

> **Private canvases**
>
> Private canvases aren't governed by CDF access management.

Charts

Capability type	Action	Scope	Description
Assets	`assets:read`	Data sets, All	Search for assets.
Time series	`timeseries:read`	Data sets, All	Search for time series.
Time series	`timeseries:write`	Data sets,	Schedule calculations.
Files	`files:read`	Data sets, All	Search for files.
Groups	`groups:list`	Current user, All	Use calculations.
Projects	`projects:list`	All	Use calculations
Sessions	`Sessions:list`	All	Monitor and schedule calculations.
Sessions	`Sessions:create`	All	Monitor and schedule calculations.
Sessions	`Sessions:delete`	All	Monitor and schedule calculations.

Configure InField

			project that InField runs on top of.
Groups	`groups:read`	Current user, All	For InField administrators to grant access to users.
3D	`3d:read`	Data sets, All	Upload 3D models to be displayed in InField.
Files	`files:write`	Data sets	Allow users to upload images.
Time series	`timeseries:write`	Data sets, Time series, Root assets, All	Allow users to upload measurement readings.

Add the InField admin users to an access group named `applications-configuration`. Learn more

Project settings

Configure default settings for how data is displayed in **search** and configure location filters to help users find and select data related to a specific physical location.

Capability type	Action	Scope	Description
Location-filters	`locationfilters:read`, `locationfilters:write`	Location-filters, All	View, create, edit, and delete location filters.
App-config	`appconfig:read`, `appconfig:write`	app-scope: Search	Create, edit, and delete search display settings for search results.

Data explorer and Image and video data

Find, validate, and learn about the data you need to build solutions in the Data explorer and Image and video management.

Annotations	`annotations:write`	All	Create new/edit existing annotations.

Data modeling

Create data models and ingest data into them.

Capability type	Action	Scope	Description
Data models	`datamodels:read`	All, Space	View data models.
Data models	`datamodels:write`	All, Space	Create, update, delete data models.
Data model instances	`datamodelinstances:read`	All, Space	View data in data models.
Data model instances	`datamodelinstances:write`	All, Space	Create, update, delete instances.
Data model instances	`datamodelinstances:write_properties`	All, Space	Write properties without allowing creation/deletion of instances.

For more details, see **Access control lists**.

Records

Manage streams and ingest or read records in the defined streams.

Capability type	Action	Scope	Description
Data models	`datamodels:write`	All, Space	Create containers
Data models	`datamodels:read`	All, Space	View containers

Streams	`streams:delete`	All	Delete streams
Records	`streamrecords:read`	All, Space	Read records from streams
Records	`streamrecords:write`	All, Space	Write records to streams

To read or write records, you must have the `datamodels:read` capability scoped to **All** or to at least the **Space** where the container is located, in addition to the stream records capabilities. For more details, see **Records and Streams**.

Data sets and Data catalog

Use the **Data sets** capability type to grant users and applications access to add or edit metadata for data sets.

To add or edit data within a data set, use the relevant resource type capability. For instance, to write time series to a data set, use the **Time series** capability type. Read more **here**.

Capability type	Action	Scope	Description
Data sets	`datasets:read`	Data sets, All	View data sets.
Data sets	`datasets:write`	Data sets, All	Create or edit data sets.

Data workflows

Use data workflows to coordinate interdependent processes.

Capability type	Action	Scope	Description
Data workflows	`workfloworchestration:read`	Data sets, All	View data workflows, versions, executions, tasks, and triggers.

Subscribe signals	`subscribesignals:read`	Current user, All	View email notification settings for workflows.
Subscribe signals	`subscribesignals:write`	Current user, All	Configure email notifications for workflow failures, timeouts, and trigger failures. Users with scope `All` can set up notifications for any user.

Diagram parsing for asset-centric data models

Find, extract, and match tags on engineering diagrams and link them to an asset hierarchy or other resource types.

Capability type	Action	Scope	Description
Files	`files:read`, `files:write`	Data sets, All	List and extract tags from engineering diagrams.
Assets	`assets:read`	All	List assets.
Events	`events:read`, `events:write`	Data sets, All	View and create annotations manually or automatically in the engineering diagrams.
Data sets	`datasets:read`	Data sets, All	List data sets.
Annotations	`annotations:write`	All	View and create annotations manually or automatically in the engineering diagrams.
Labels	`labels:read`, `labels:write`	All	View the approval status of parsed engineering diagrams.

Diagram parsing for data modeling

Diagram parsing	`diagramParsing:read`	No scope	See the results of completed parsing jobs and read the outputs from parsed files.
Data model instances	`dataModelInstances:read`	Space IDs, All	See the results of completed parsing jobs and read the outputs from parsed files.
Diagram parsing	`diagramParsing:read`, `diagramParsing:write`	No scope	Start a diagram parsing job.
Data model instances	`dataModelInstances:read`, `dataModelInstances:write`	Space IDs, All	Start a diagram parsing job.
Sessions	`sessions:create`	All	Start a diagram parsing job.
Location filters	`locationFilters:read`	Location filters, All	List files belonging to a location.

Document parser

Extract data from documents, such as datasheets, equipment specifications, or process flow diagrams.

Data models	`datamodels:write`	Space IDs, All	Approve the results of completed parsing jobs. The data are saved in a data model instance.
Files	`files:read`	Data sets, All	Start a document parsing job or see the results of completed parsing jobs.

Extraction pipelines

Set up and monitor extraction pipelines and report the pipeline run history.

User	Action	Capability	Description
End-user	Create and edit extraction pipelines	`extractionpipelines:write`	Gives access to create and edit individual pipelines and edit notification settings. Ensure that the pipeline has `read` access to the data set being used by the extraction pipeline.
	View extraction pipelines	`extractionpipelines:read`	Gives access to list and view pipeline metadata.
	Create and edit extraction configurations	`extractionconfigs:write`	Gives access to create and edit an extractor configuration in an extraction pipeline.
	View extraction configurations	`extractionconfigs:read`	Gives access to view an extractor configuration in an extraction pipeline.
	View extraction logs	`extractionruns:read`	Gives access to view run history reported by the extraction pipeline runs.
Extractor	Read extraction configurations	`extractionconfigs:read`	Gives access to read an extractor configuration from an extraction pipeline.
	Post extraction logs	`extractionruns:write`	Gives access to post run history reported by the extraction pipeline runs.
Third-party actors	Create and edit extraction pipelines	`extractionpipelines:write`	Gives access to create and edit individual pipelines and edit notification settings. Ensure that the pipeline has `read` access to the data set being used by the extraction pipeline.
	Create and edit extraction configurations	`extractionconfigs:write`	Gives access to create and edit the extractor configuration from an extraction pipeline.

Capability type	Action	Scope	Description
Functions	`functions:write`	All	Create, call and schedule functions.
Functions	`functions:read`	All	Retrieve and list functions, retrieve function responses and function logs.
Files	`files:read`	All	View functions.
Files	`files:write`	All	Create functions.
Sessions	`sessions:create`	All	Call and schedule functions.

Hosted extractors

Cognite Event Hub extractor

Capability type	Action	Scope	Description
Hosted extractors	`hostedextractors:read`, `hostedextractors:write`	All	Create an Event Hub extractor.
Time series	`timeseries:read`, `timeseries:write`	Data sets, Time series, Root assets, All	Write data points and time series.
Events	`event:read`, `events:write`	Data sets, All	Write events.
RAW	`raw:read`, `raw:write`	Tables, All	Write RAW rows.
Data models	`datamodelinstances:write`	Space IDs, All	Write to data models.

Cognite Kafka extractor

Time series	`timeseries:read`, `timeseries:write`	Data sets, Time series, Root assets, All	Write data points and time series.
Events	`event:read`, `events:write`, `assets:read`	Data sets, All	Write events.
RAW	`raw:read`, `raw:write`	Tables, All	Write RAW rows.
Data models	`datamodelinstances:write`	Space IDs, All	Write to data models.

Cognite MQTT extractor

Capability type	Action	Scope	Description
Hosted extractors	`hostedextractors:read`, `hostedextractors:write`	All	Create an MQTT extractor.
Time series	`timeseries:read`, `timeseries:write`	Data sets, Time series, Root assets, All	Write data points and time series.
Events	`event:read`, `events:write`, `assets:read`	Data sets, All	Write events.
RAW	`raw:read`, `raw:write`	Tables, All	Write RAW rows.
Data models	`datamodelinstances:write`	Space IDs, All	Write to data models.

Cognite REST extractor

Capability type	Action	Scope	Description
Hosted extractors	`hostedextractors:read`, `hostedextractors:write`	All	Create a REST extractor.

	`events:write`, `assets:read`		
RAW	`raw:read`, `raw:write`	Tables, All	Write RAW rows.
Data models	`datamodelinstances:write`	Space IDs, All	Write to data models.

Match entities

Create and tune models to automatically contextualize resources.

Capability type	Action	Scope	Description
Entity matching	`entitymatchingAcl:read`	All	List and view entity matching models.
Entity matching	`entitymatchingAcl:write`	All	Create, update, delete entity matching models.
Assets	`assets:read`	Data sets, All	Match entities to assets.

On-premises extractors

> The service principal used by the on-premises extractors must also be assigned the capabilities listed in the **Feature capabilities table**.

DB extractor

Extract data from any database supporting Open Database Connectivity (ODBC) drivers.

Time series	`timeseries:read`, `timeseries:write`	Data sets, All	Use when query destination is set to `time_series`.
Events	`events:read`, `events:write`	Data sets, All	Use when query destination is set to `events`.
Files	`files:read`, `files:write`	Data sets, All	Use when query destination is set to `files`.
Sequences	`sequences:read`, `sequences:write`	Data sets, All	Use when query destination is set to `sequences`.
Extraction pipeline runs	`extractionruns:write`	Data sets, Extraction pipelines, All	Allow the extractor to report state and heartbeat back to CDF.
Remote configuration files	`extractionconfigs:read`	Data sets, Extraction pipelines, All	Use versioned extractor configuration files stored in the cloud.

Documentum extractor

Extract documents from OpenText Documentum or OpenText D2 systems.

RAW	`raw:read`, `raw:write`, `raw:list`	Tables, All	Ingest metadata to CDF RAW.
Extraction pipeline runs	`extractionruns:write`	Data sets, Extraction pipelines, All	Allow the extractor to report state and heartbeat back to CDF.
Remote configuration files	`extractionconfigs:read`	Data sets, Extraction pipelines, All	Use versioned extractor configuration files stored in the cloud.

EDM extractor

Connect to the **Landmark Engineers Data Model** server and extract data through the **Open Data protocol** (OData) from DecisionSpace Integration Server (DSIS) to **CDF RAW**.

Capability type	Action	Scope	Description
RAW	`raw:read`, `raw:write`, `raw:list`	Tables, All	Ingest data from Landmark EDM model into CDF RAW.

File extractor

Connect to local file systems, SharePoint Online Document libraries, and network sharing protocols, like FTP, FTPS, and SFTP, and extract files into CDF.

Capability type	Action	Scope	Description
Files	`files:read`, `files:write`	Data sets, All	Ingest files.
RAW	`raw:read`, `raw:write`, `raw:list`	Tables, All	For state store configured to use CDF RAW.
Data models	`datamodelinstances:read`,	Space IDs, All	Use when `data_model` is set as destination

			back to CDF.
Remote configuration files	`extractionconfigs:read`	Data sets, Extraction pipelines, All	Use versioned extractor configuration files stored in the cloud.

OPC classic extractor

Capability type	Action	Scope	Description
Time series	`timeseries:read`, `timeseries:write`	Data sets, All	Ingest time series
Extraction pipelines	`extractionpipelines:read`, `extractionruns:write`, `extractionconfigs:read`	Data sets, All	For extraction pipelines and remote configuration.
Assets	`assets:read`, `assets:write` if `metadata-sinks.clean.assets` is set	Data sets, All	Ingest assets
RAW	`raw:read`, `raw:write`, `raw:list` if `state-store` is set to use CDF RAW	Tables, All	
Data sets	`datasets:read`, if using `dataset-external-id`	Data sets, All	

OPC UA extractor

Extract time series, events, and asset data via the OPC UA protocol.

Capability type	Action	Scope	Description
Time series	`timeseries:read`, `timeseries:write`	Data sets, All	Ingest time series.

Events	`events:read`, `events:write`	Data sets, All	Ingest events if enabled.
RAW	`RAW:read`, `RAW:write`, `RAW:list`	Tables, All	Ingest metadata to CDF RAW or the state-store is set to use CDF RAW.
Relationships	`relationships:read`, `relationships:write`	Data sets, All	Ingest relationships if enabled in the configuration.
Data sets	`data-sets:read`	Data sets, All	Ingest the data set external ID if enabled in the configuration.
Extraction pipeline runs	`extractionruns:write`	Data sets, Extraction pipelines, All	Allow the extractor to report state and heartbeat back to CDF.
Extraction pipeline runs	`extractionruns:write`	Data sets, Extraction pipelines, All	Allow the extractor to report state and heartbeat back to CDF.
Remote configuration files	`extractionconfigs:read`	Data sets, Extraction pipelines, All	Use versioned extractor configuration files stored in the cloud.
Data model instances	`datamodelinstances:read`, `datamodelinstances:write`	Space IDs, All	Use when writing to Core Data Models.
Data models	`datamodel:read`, `datamodel:write`	Space IDs, All	Use when creating data models from OPC-UA extractor (alpha feature)
Streams	`streamrecords:read`, `streamrecords:write`, `streams:read`, and `streams:write`	Space IDs, All	Use when ingesting to CDF streams (alpha feature)

RAW	`raw:read`, `raw:write`, `raw:list`	Tables, All	Ingest configurable OSDU records, such as wells, wellbores, and well logs.
Files	`files:read`, `files:write`	Data sets, All	Ingest linked data files.
Extraction pipelines	`extractionpipelines:read`, `extractionpipelines:write`	Data sets, Extraction pipelines, All	Create and edit extraction pipelines.
Extraction pipeline runs	`extractionruns:read`, `extractionruns:write`	Data sets, Extraction pipelines, All	Allow the extractor to report state and heartbeat back to CDF.

PI AF extractor

Extract data from the OSIsoft PI Asset Framework (PI AF).

Capability type	Action	Scope	Description
RAW	`raw:read`, `raw:write`, `raw:list`	Tables, All	Ingest to CDF RAW and for state store configured to use CDF RAW.
Extraction pipeline runs	`extractionruns:write`	Data sets, Extraction pipelines, All	Allow the extractor to report state and heartbeat back to CDF.
Remote configuration files	`extractionconfigs:read`	Data sets, Extraction pipelines, All	Use versioned extractor configuration files stored in the cloud.

PI extractor

Extract time series data from the OSISoft PI Data Archive.

RAW	`raw:read` , `raw:write` , `raw:list`	Tables, All	Ingest to CDF RAW and for state store configured to use CDF RAW.
Events	`events:read` , `events:write`	Data sets, All	Log extractor incidents as events in CDF.
Extraction pipeline runs	`extractionruns:write`	Data sets, Extraction pipelines, All	Allow the extractor to report state and heartbeat back to CDF.
Remote configuration files	`extractionconfigs:read`	Data sets, Extraction pipelines, All	Use versioned extractor configuration files stored in the cloud.
Data models	`datamodelinstances:read` , `datamodelinstances:write`	Space IDs, All	Use when writing to Core Data Models.

PI Replace utility

Re-ingest time series to CDF by optionally deleting a data point time range and ingesting the data points in PI for that time range.

Capability type	Action	Scope	Description
Time series	`timeseries:read` , `timeseries:write`	Data sets, All	Re-ingest time series into CDF.
RAW	`RAW:read` , `RAW:write` , `RAW:list`	Tables, All	Ingest to CDF RAW and for state store configured to use CDF RAW.
Events	`events:read` , `events:write`	Data sets, All	Log extractor incidents as events in CDF.
Extraction pipeline runs	`extractionruns:write`	Data sets, Extraction pipelines, All	Allow the extractor to report state and heartbeat back to CDF.

Data models	`datamodelinstances:read`, `datamodelinstances:write`	Space IDs, All	Use when writing to Core Data Models.

SAP extractor

Extract data from SAP ERP servers into Cognite Data Fusion.

Capability type	Action	Scope	Description
RAW	`RAW:read`, `RAW:write`, `RAW:list`	Tables, All	Ingest to CDF RAW and for state store configured to use CDF RAW.
Time series	`timeseries:read`, `timeseries:write`	Data sets, All	Use when destination is set to `time_series`.
Events	`events:read`, `events:write`	Data sets, All	Use when destination is set to `events`.
Files	`files:read`, `files:write`	Data sets, All	Use when extracting `attachments` from SAP to CDF files.
Extraction pipeline runs	`extractionruns:write`	Data sets, Extraction pipelines, All	Allow the extractor to report state and heartbeat back to CDF.
Remote configuration files	`extractionconfigs:read`	Data sets, Extraction pipelines, All	Use versioned extractor configuration files stored in the cloud.

Studio for Petrel extractor

Connect to SLB Studio for Petrel through the Ocean SDK and stream Petrel object data to the CDF files service as protobuf objects.

WITSML extractor

Connect via the Simple Object Access Protocol (SOAP) and the Energistics Transfer Protocol (ETP) and extract data using the Wellsite Information Transfer Standard Markup Language (WITSML) into CDF.

Capability type	Action	Scope	Description
Time series	`timeseries:read`, `timeseries:write`, `timeseries:list`	Data sets, All	Ingest WITSML **growing objects**, such as WITSML logs, into the CDF time series services.
Sequences	`sequences:read`, `sequences:write`, `sequences:list`	Data sets, All	Ingest WITSML **growing objects**, such as WITSML logs, into the CDF sequences services.
RAW	`raw:read`, `raw:write`, and `raw:list`	Tables, All	Ingest WITSML **non-growing objects**, such as `wellbore`, into CDF RAW.
Files	`files:read` and `files:write`	Data sets, All	Ingest WITSML SOAP queries and responses to CDF files.
Extraction pipelines	`extractionpipelines:write`, `extractionpipelines:read`	Data sets, Extraction pipelines, All	Allow the extractor to report state and heartbeat back to CDF.
Extraction pipeline runs	`extractionruns:write`	Data sets, Extraction pipelines, All	Allow the extractor to report state and heartbeat back to CDF.

PostgreSQL gateway

			for the CDF resources you want to ingest data into. For instance, to ingest **assets**, add `assets:read` and `assets:write`.

> If you revoke the capabilities in the CDF group, you also revoke access for the PostgreSQL gateway.

Simulator connectors

Integrate existing simulators with CDF to remotely run simulations.

Capability type	Action	Scope	Description
Simulators	`simulators:manage`	Data sets, All	Connect to the simulators API.
Time series	`timeseries:read`, `timeseries:write`	Data sets, All	Use CDF time series as source/destination for simulation data.
Extraction pipelines	`extractionpipelines:read`, `extractionpipelines:write`	Data sets, All	Store the connector configuration remotely.
Files	`files:read`	Data sets, All	Download simulation model files.
Data sets	`datasets:read`	Data sets, All	View data sets.

Simulators users

Set access to the user interface for integrating existing simulators with CDF to remotely run simulations.

	`simulators:run`		
Time series	`timeseries:read` ,	Data sets, All	Use CDF time series as source/destination for simulation data.
Files	`files:read` , `files:write`	Data sets, All	Upload and download simulation model files.
Data sets	`datasets:read`	Data sets, All	View data sets.

Staged data

Work with tables and databases in CDF RAW.

Capability type	Action	Scope	Description
RAW	`RAW:read` , `RAW:list`	Tables, All	View tables and databases in CDF RAW.
RAW	`RAW:write`	Tables, All	Create, update, delete tables and databases in CDF RAW.

Streamlit apps

Build custom web applications in CDF.

Capability type	Action	Scope	Description
Files	`files:read` , `files:write`	Data sets, All	Edit Streamlit applications.
Files	`files:read`	Data sets, All	Use Streamlit applications.

Resource types	`read` and `write` actions according to the CDF resources you want to read from and write to using transformations.	Data sets, All	For instance, to transform data in CDF RAW and write data to assets, add `RAW:read` and `assets:write`.
Transformations	`transformations:read`	All	View transformations.
Transformations	`transformations:write`	All	Create, update, delete CDF transformations.
Session	`sessions:create`	All	Run scheduled transformations. You must also set the `token_url`. Read more **here**.

> **Legacy access to transformations via group name**
>
> To ensure backward compatibility, groups named `transformations` or `jetfire` are treated as having both `transformations:read:All` and `transformations:write:All`. We're deprecating this access control method and will remove it in a future release.

Upload 3D models

Upload and work with 3D models, 3D revisions and 3D files.

Capability type	Action	Scope	Description
3D	`3d:read`, `file:read`	Data sets, All	View 3D models.
3D	`3d:create`, `file:read`	Data sets, All	Upload 3D models to CDF.
3D	`3d:update`	Data sets, All	Update existing 3D models in CDF.
3D	`3d:delete`	Data sets, All	Delete 3D models.

Data sets (Optional)	`dataset:read` , `dataset:owner`	Data sets, All	Read and write the dataset scoped source file, 3D model, and its outputs.
Data model instances	`datamodelinstances:read`	Space IDs, All	View 360° image data stored in data models.
Data model instances	`datamodelinstances:write`	Space IDs, All	Create, update, delete data model instances for 360° images.
Data models	`datamodel:read` , `datamodel:write`	Space IDs, All	Upload 360° image metadata to data model views.
Sessions	`sessions:create`	All	Start 360° image upload/processing.

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